

Python with MariaDB

Running a few Database Programs

Prerequisites for This Lesson

- Basic Python 3.x
  - Variables
  - Sequence structures
  - Selections structures
  - Repetitive structures
  - Functions
  - Modules
  - packages
- Basic Database knowledge
  - What is a DBMS?
  - What is a database?
  - What is MariaDB?
  - What is PhpMyAdmin?
    - http://localhost/phpmyadmin/
  - Create a database
  - Create a table
  - **Write simple SQL queries**

Step 1 – Create a Database with Data

Some Simple SQL Queries

- SHOW DATABASES
- USE <database>
- SELECT * FROM <table>
- INSERT INTO tbl_mytable (No, FName, MName, City) VALUES (3, 'John', 'Doe', 'Kandy');
- SELECT No, FName FROM tbl_mytable
- SELECT * FROM tbl_mytable WHERE No = 2;
- DELETE FROM tbl_mytable WHERE No = 3

Create a Database with Below Specification

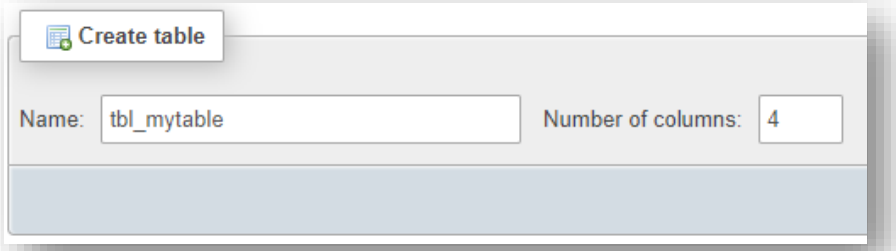
- Database Name – db_test_python
- Table name – tbl_mytable
- Columns – 4
  - Column 1 – No (INT)
  - Column 2 – FName (VARCHAR size 15)
  - Column 3 – MName (VARCHAR size 15)
  - Column 4 – City (VARCHAR size 15)

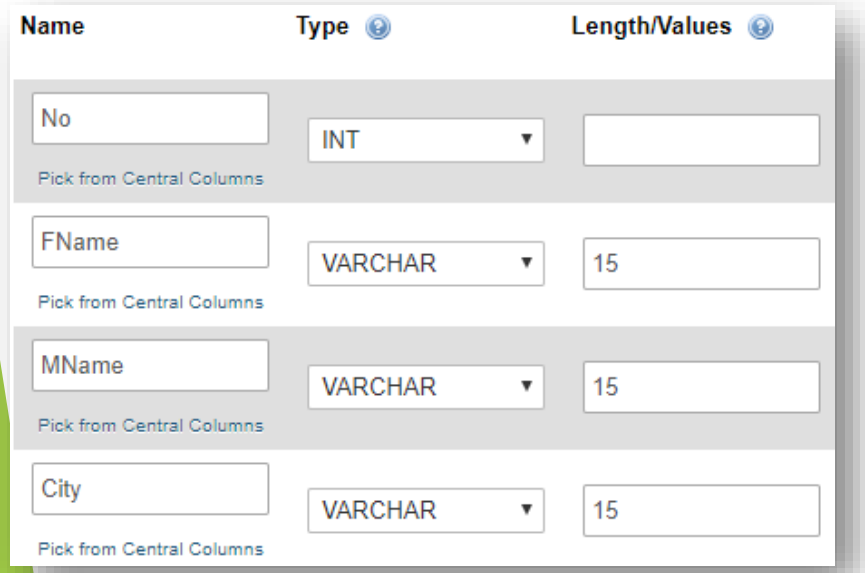
Load The Table With Sample Data

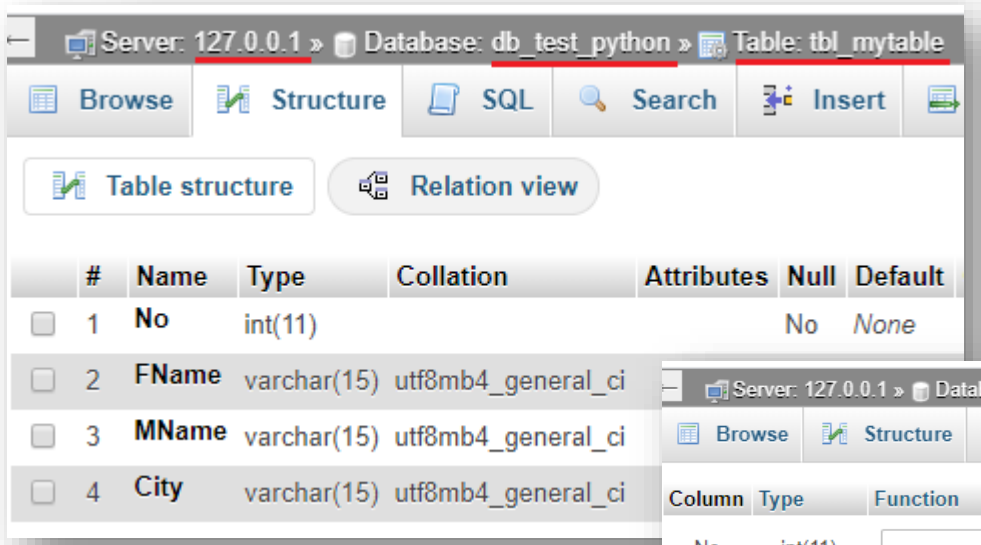
No	FName	MName	City
1	Nishan	Saliya	Gampaha
2	Janaka	Sukith	Maradana
3	John	Doe	Kandy
4	Lie	Kun	Wuhan

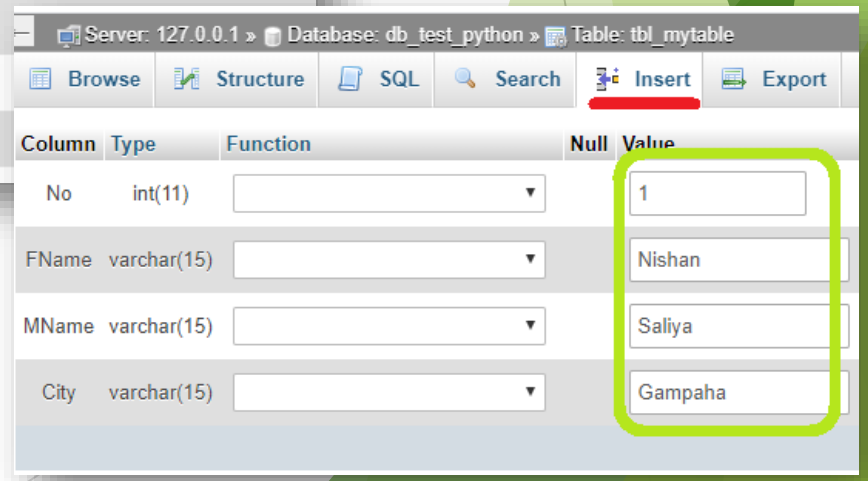
Some Random Images as Aid











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Step 2 – Write Python Codes

Testing The Connection

- Try the below 2 commands in **Python IDLE** to see the result

```
import mariadb
mariadb.connect(host="localhost",user="root",password="")
```

```
>>> import mariadb
>>> mariadb.connect(host='localhost',user='root',password='')
<mariadb.connection connected to 'localhost' at 000001E12E011010>
>>>
```

Database Operations for Today

- Test connectivity
- View databases
- Display table data
- Insert data
- Delete data
- Update data

Given code samples does not use Exception handling! It's **highly recommended** to use it in real programs!

A sample is given for the connection testing program.

Check Connectivity

```
 1 import mariadb
 2 from mariadb import Error
 3
 4 try:
 5     conn = mariadb.connect(host='localhost',user='root',password='')
 6     conn.ping()
 7     print("Connected")
 8 except Error as e:
 9     print("Oops!")
10     print(e)
11
```

View The Available Databases

```
1  import mariadb
2
3  # Open database connection
4  db = mariadb.connect(host="localhost",user="root",password="")
5
6  # prepare a cursor object using cursor() method
7  cursor = db.cursor()
8
9  # execute SQL query using execute() method.
10 cursor.execute("SHOW DATABASES")
11
12 # Fetch results using fetchall() method.
13 data = cursor.fetchall()
14 print ("Available databases : %s " % data)
15
16 # disconnect from server
17 db.close()
18
```

View The Available Databases in a Loop

```
1  import mariadb
2
3  # Open database connection
4  db = mariadb.connect(host="localhost",user="root",password="")
5
6  # prepare a cursor object using cursor() method
7  cursor = db.cursor()
8
9  # execute SQL query using execute() method.
10 cursor.execute("SHOW DATABASES")
11
12 # Fetch results using fetchall() method.
13 data = cursor.fetchall()
14
15 for item in data:
16     print(item)
17
18 # disconnect from server
19 db.close()
20
```

Display Information from a Table (All)

```
1  import mariadb
2
3  # Open database connection with a dictionery
4
5  conDict = {'host':'localhost',
6             'database':'db_test_python',
7             'user':'root',
8             'password':''}
9
10 db = mariadb.connect(**conDict)
11
12 # prepare a cursor object using cursor() method
13 cursor = db.cursor()
14
15 # execute SQL query using execute() method.
16 cursor.execute("SELECT * FROM tbl_mytable")
17
18 # Fetch results using fetchall() method.
19 data = cursor.fetchall()
20
21 for item in data:
22     print(item)
23
24 # disconnect from server
25 db.close()
26
```

Display Information from a Table (Selected)

```
1  import mariadb
2
3  # Open database connection with a dictionery
4
5  conDict = {'host':'localhost',
6             'database':'db_test_python',
7             'user':'root',
8             'password':''}
9
10 db = mariadb.connect(**conDict)
11
12 # prepare a cursor object using cursor() method
13 cursor = db.cursor()
14
15 # execute SQL query using execute() method.
16 cursor.execute("SELECT No, FName, City FROM tbl_mytable")
17
18 # Fetch results using fetchall() method.
19 data = cursor.fetchall()
20
21 for item in data:
22     print(item)
23
24 # disconnect from server
25 db.close()
26
```

Display Information from a Table (WHERE)

```
1  import mariadb
2
3  # Open database connection with a dictionery
4
5  conDict = {'host':'localhost',
6             'database':'db_test_python',
7             'user':'root',
8             'password':''}
9
10 db = mariadb.connect(**conDict)
11
12 # prepare a cursor object using cursor() method
13 cursor = db.cursor()
14
15 # execute SQL query using execute() method.
16 cursor.execute("SELECT * FROM tbl_mytable WHERE No = 3")
17
18 # Fetch results using fetchall() method.
19 data = cursor.fetchall()
20
21 for item in data:
22     print(item)
23
24 # disconnect from server
25 db.close()
26
```

Insert Records to Table

```
1  import mariadb
2
3  # Open database connection with a dictionery
4
5  conDict = {'host':'localhost',
6             'database':'db_test_python',
7             'user':'root',
8             'password':''}
9
10 db = mariadb.connect(**conDict)
11
12 # prepare a cursor object using cursor() method
13 cursor = db.cursor()
14
15 # execute SQL query using execute() method.
16 myInsertText = "INSERT INTO tbl_mytable VALUES (6,'Judi','Zilva','Rathmalana')"
17 cursor.execute(myInsertText)
18
19 db.commit()
20 db.close()
21
```

Insert Records to Table 2 (User Input)

```
1  import mariadb
2
3  # Open database connection with a dictionery
4  conDict = {'host':'localhost',
5             'database':'db_test_python',
6             'user':'root',
7             'password':''}
8
9  db = mariadb.connect(**conDict)
10
11 # prepare a cursor object using cursor() method
12 cursor = db.cursor()
13
14 # Read user input
15 uNo = int(input("Type user number : "))
16 uFName = input("Type first name : ")
17 uMName = input("Type middle name : ")
18 uCity = input("Type city : ")
19
```

Insert Records to Table 2 (User Input) – P2

```
19
20 # execute SQL query using execute() method.
21 mySQLText = "INSERT INTO tbl_mytable (No, FName, MName, City) VALUES (%s,%s,%s,%s)"
22 myValues = (uNo, uFName, uMName, uCity)
23 cursor.execute(mySQLText, myValues)
24
25 # Commit the change
26 db.commit()
27 db.close()
```

Delete A Record

```
1  import mariadb
2
3  # Open database connection with a dictionery
4  conDict = {'host':'localhost',
5             'database':'db_test_python',
6             'user':'root',
7             'password':''}
8
9  db = mariadb.connect(**conDict)
10
11 # prepare a cursor object using cursor() method
12 cursor = db.cursor()
13
14 # execute SQL query using execute() method.
15 cursor.execute("DELETE FROM tbl_mytable WHERE No = 7")
16
17 # Commit the change
18 db.commit()
19
20 # disconnect from server
21 db.close()
22
```

Delete A Given Record

```
1  import mariadb
2
3  # Open database connection with a dictionery
4  conDict = {'host':'localhost',
5             'database':'db_test_python',
6             'user':'root',
7             'password':''}
8
9  db = mariadb.connect(**conDict)
10
11 # prepare a cursor object using cursor() method
12 cursor = db.cursor()
13
14 uNo = input("Type a user number to delete : ")
15
16 # execute SQL query using execute() method.
17 cursor.execute("DELETE FROM tbl_mytable WHERE No = " + uNo + "")
18
19 # Commit the change
20 db.commit()
21 # disconnect from server
22 db.close()
23
```

Update A Record

```
1  import mariadb
2
3  # Open database connection with a dictionery
4  conDict = {'host':'localhost',
5             'database':'db_test_python',
6             'user':'root',
7             'password':''}
8
9  db = mariadb.connect(**conDict)
10
11 # prepare a cursor object using cursor() method
12 cursor = db.cursor()
13
14 # execute SQL query using execute() method.
15 updTxt = "UPDATE tbl_mytable SET MName = 'Malaka' WHERE No = 2"
16 cursor.execute(updTxt)
17
18 # Commit the change
19 db.commit()
20
21 # disconnect from server
22 db.close()
23
```

Update A Given Record

```
1  import mariadb
2
3  # Open database connection with a dictionery
4  conDict = {'host':'localhost',
5             'database':'db_test_python',
6             'user':'root',
7             'password':''}
8
9  db = mariadb.connect(**conDict)
10
11 # prepare a cursor object using cursor() method
12 cursor = db.cursor()
13
14 uNo = input("Type a user number : ")
15 uFName = input ("Type new first name : ")
16
17 # execute SQL query using execute() method.
18 updTxt = "UPDATE tbl_mytable SET FName = '" + uFName + "' WHERE No = " + uNo
19 cursor.execute(updTxt)
20
21 # Commit the change
22 db.commit()
23 db.close()
24
```